Worksheet 7

Integrated cases – EXERCISES

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Integrated case A – Continuous assessment preparation

A school surveys 200 students about the maximum monthly price they would pay for an extended catering offer.

Price (euros)	$[5, 10[$	$[10, 15[$	$[15, 20[$	$[20, 30[$	$[30, 40[$	$[40, 60[$
Count	10	25	45	60	40	20

Identify the observed population, statistical unit, variable, and type; build the complete table (widths, densities, relative frequencies, ICC, DCC, midpoints); draw the histogram and cumulative curves; calculate modal class, median, mean, Q_1, Q_3, variance, standard deviation, and coefficient of variation; find the price affordable to about 65% of students; then discuss a satisfied/not satisfied qualitative variable and suitable graph.

Integrated case B – Final exam preparation

Part 1 – Regression. For $x = (4, 5, 7, 8, 10, 11, 13, 15)$ hundreds of customers and $y = (8, 10, 14, 15, 20, 22, 26, 31)$ tens of thousands of euros, calculate R, the line $\hat{y} = ax + b$, and forecast y for $x = 18$.

Part 2 – Indices. For products A, B, C with (p_0, q_0, p_1, q_1) equal to $(10, 100, 12, 110)$, $(20, 50, 22, 48)$, and $(30, 40, 33, 45)$, calculate Laspeyres, Paasche, Fisher price/quantity indices and the value index.

Part 3 – Quarterly series. Trend is $T_t = 60+2t$ and corrected seasonal coefficients are $0.80, 1.10, 0.90, 1.20$ for Q1–Q4. Check normalisation and forecast $t = 13$ to 16 under a multiplicative model.